

Metal Bath – Heating & Cooling Instrument

Software Operation Manual

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1 Overview

1.1 Purpose of This Manual

The purpose of this manual is to enable users to correctly, safely, and effectively operate the Metal Bath Heating & Cooling Control System to achieve precise temperature control, meeting laboratory and industrial application requirements.

1.2 Icon Definitions

Icon	Meaning
Manual Settings (highlighted)	
Program Control (highlighted)	
Program Edit (highlighted)	
Run Curve (highlighted)	
System Settings (highlighted)	
Manual Settings (normal)	
Program Control (normal)	
Program Edit (normal)	
Run Curve (normal)	
System Settings (normal)	

Manual Settings (disabled)	
Program Control (disabled)	
Program Edit (disabled)	
Run Curve (disabled)	
System Settings (disabled)	
Information prompt	Please pay attention before operating
Run button (highlighted / normal / disabled)	
Pause button (highlighted / normal / disabled)	
Stop button (highlighted / normal / disabled)	
Common temperature: 4°C (normal / highlighted)	
Common temperature: 37°C (normal / highlighted)	
Common temperature: 65°C (normal / highlighted)	
Common temperature: 85°C (normal / highlighted)	
Restore room temperature (normal / highlighted)	
Rate mode: Basic, High-speed, Extreme	
Module model	
Basic / High-speed / Extreme mode (normal & highlighted)	
Confirm (normal)	
View (normal / highlighted)	
Delete (normal / highlighted)	
Previous page / Next page (normal)	
Back (normal)	
Rename (normal)	
Edit (normal)	
Program add / Program delete (normal)	
Low-temperature preservation (normal / highlighted)	
Settings (normal / highlighted)	
Power-on preset (normal / highlighted)	
Data export (normal / highlighted)	
Self-test (normal / highlighted)	

Log (normal / highlighted)	
Chinese/English (normal / highlighted)	
Hot lid mode (normal / highlighted)	
Alarm info (normal / highlighted)	
Calibration (normal / highlighted)	
Low-temp preservation / Power-on preset / Hot lid / Touch-screen prompt tone - ON (highlighted)	
Low-temp preservation / Power-on preset / Hot lid / Touch-screen prompt tone - OFF (disabled)	
Log details (disabled / highlighted)	
Clear (normal)	

2 System Login

After the device is powered on, the screen first displays the boot screen (about 3 seconds), then automatically enters the power-on self-test process. The system will check whether the temperature sensor is working properly within 10 seconds. After the self-test passes, it automatically enters the Manual Settings page.

Figure 2-1 Power-on Self-test Page

After the self-test passes, as shown in the figure below, it enters the main Manual Settings page and is ready for operation.

Figure 2-2 Manual Settings Main Page

3 Manual Settings

3.1 Manual Settings Page Overview

The Manual Settings page is the device's main operation interface. Users can complete temperature setting, time setting, rate mode selection, and start/stop control on this page.

Figure 3-1 Manual Settings Page

- **Module model:** Displays the currently identified module model (automatically identified via hardware ID pin, supporting up to 16 module types).
- **Rate mode:** Three heating/cooling rate levels can be selected: Basic, High-speed, and Extreme.
- **Common temperatures:** Provides four common-temperature shortcut buttons: 4°C, 37°C, 65°C, and 85°C. Click to quickly set the target temperature.
- **Restore room temperature:** One-click to restore the sample temperature to 23°C. When the temperature reaches 23±1°C, it stops automatically, the button pops up, and the buzzer beeps once.
- **Set temperature:** Click the input box to set the target temperature, range 4~100°C, accuracy 0.1°C.
- **Set time:** Set the hold time (HH:MM:SS), maximum 99:59:59.

- **Real-time temperature:** Displays the current real-time temperature of the sample area.
- **Hold countdown:** After the temperature reaches the set value, the countdown starts automatically.
- **Run time:** Displays the device's cumulative running time.
- **Run/Pause/Stop:** Three control buttons to control the device start/stop.

Note: Both set temperature and set time have a power-loss save function. After a power outage and restart, the previous set values will be restored.

3.2 Temperature Setting

Click the "Set Temperature" input box to bring up the numeric keypad. Enter the target temperature value and click "Confirm". The temperature range is 0~105°C; a prompt will appear if the value is out of range.

Figure 3-2 Temperature Input Keypad

3.3 Time Setting

Click the "Set Time" input box to bring up the time setting interface, where you set hours, minutes, and seconds separately. The maximum value is 99:59:59. When the set time is 0, the device will run continuously (no countdown).

3.4 Rate Mode

Click the "Rate Mode" drop-down box to select one of the following three heating/cooling rates:

Mode	Heating rate	Cooling rate	Applicable scenario
Basic	~2°C/min	~0.5°C/min	Experiments with low requirements for temperature change rate
High-speed	~5°C/min	~1°C/min	Experiments requiring fast arrival at target temperature
Extreme	Unlimited	Unlimited	Reaching the target temperature at the fastest speed

3.5 Common Temperature Shortcuts

The Manual Settings page provides four common-temperature shortcut buttons: 4°C, 37°C, 65°C, and 85°C. Click the corresponding button to automatically set the target temperature to that value without manual input. On the Manual Settings page, when the set temperature is not a common temperature, the common-temperature buttons pop up.

3.6 Restore Room Temperature

Click the "Restore Room Temperature" button, and the device will automatically control the temperature to 23°C. When the temperature reaches 23±1°C, it stops automatically, the button

pops up, and the buzzer beeps once. This function is suitable for safely removing samples after an experiment.

During the Restore Room Temperature operation, clicking Stop can cancel the restore room temperature function. When stopped, the fan will continue to run for 10 seconds before stopping.

4 Program Control

4.1 Program Control Page Overview

The Program Control page is used to execute preset multi-step temperature control programs, enabling automated program-controlled operation.

Figure 4-1 Program Control Page

- **Set temperature:** Displays the target temperature of the current step (cannot be modified directly; set in the Program Edit page).
- **Set time:** Displays the hold time of the current step (HH:MM:SS).
- **Rate mode:** Displays the heating/cooling rate mode of the current step.
- **Program name:** The drop-down box in the upper right corner selects the program to run (can also be selected in the Program Edit page).
- **Current step:** Displays the step number currently being executed.
- **Real-time temperature:** Displays the current real-time temperature of the sample area.
- **Hold countdown:** The countdown after the current step temperature is reached.
- **Run time:** Displays the cumulative run time in program control mode.
- **Run/Pause/Stop:** Controls the program start/stop.

Note: In program control mode, after the current step countdown ends, it automatically switches to the next step. After all steps are completed, it stops automatically and emits a buzzer prompt.

4.2 Program Edit

4.2.1 Program Library Page

Figure 4-2 Program Edit Page (Program Library)

On the Program Edit page, you can select the program to run and view detailed program steps. Up to 5 programs are supported, and each program can contain up to 10 steps. Configured programs have a power-loss save function; after a power outage and restart, the previous set values will be restored.

- Click the "View" button to view the detailed content of the current program.
- Click the "Delete" button to delete the corresponding program.

4.2.2 Program View

After clicking the "View" button, the Program View page is entered, displaying all step details of that program (target temperature, hold time, rate mode). Click the "Program Edit" button to jump to the Program Edit page to edit the current program's content. The current program's name can be changed via the drop-down box.

Figure 4-3 Program Edit - View Page (Program Library)

4.2.3 Program Edit Page

Figure 4-4 Program Edit - Edit Page (Program Library)

1. On the left side of the page there are 3 plus-sign buttons. When a button is pressed, the text box in the corresponding row becomes editable. When a button pops up, the content of the corresponding row's text box is deleted, and subsequent steps automatically move forward. (For the 10 steps, they can only be added in sequence; when the previous button is not pressed, the next button is not touchable.)
2. Steps are paginated via "Previous Page" and "Next Page". A program can store up to 10 steps.
3. After editing is complete, click the "Save" button to successfully save all step settings of the current program. If time or temperature is 0, it cannot be saved and a pop-up reminder appears. A pop-up reminder also appears on successful save. The temperature must be within 4~100°C and the set time cannot be 0, otherwise a pop-up prompt appears.

Note: Click the "Cancel" or "Back" button to return to the Program Library page.

5 Run Curve

Figure 5-1 Run Curve Page

The Run Curve page displays the temperature change trend in real time, helping users intuitively understand the temperature control process.

- The vertical axis is temperature (°C), and the horizontal axis is time (min).
- The right side marks common temperature reference lines: 4°C, 37°C, 65°C, 85°C, 100°C.
- Up to 12,800 data points can be recorded; each page displays about 1 hour of data (720 points).
- Use the Page X / Previous / Next buttons to page through historical temperature trends, supporting quick jumps.
- The curve updates in real time during operation; after stopping, historical data can be reviewed.

Note: Paging via the Page X / Previous / Next buttons is only possible after the current page has finished drawing.

6 System Settings

6.1 System Settings Page Overview

Figure 6-1 System Settings Page

1. When the system settings are clicked during manual settings or program-controlled operation, all spheres except Settings and Calibration are disabled.
2. After entering System Settings → Settings and Calibration menu, the Manual Settings and Program Edit menus need to be grayed out.

The System Settings page uses a circular icon layout and contains the following sub-function modules:

No.	Module	Description
1	Low-temperature preservation	Automatically enters low-temperature preservation mode after the run ends
2	Power-on preset	Automatically pre-heats/pre-cools to the set temperature after power-on
3	Self-test	Re-executes the power-on self-test process
4	Chinese/English	Interface language switching (not enabled in current version)
5	Hot lid mode	Independently controls the hot lid temperature
6	Alarm information	View historical alarm records
7	Calibration	Temperature compensation calibration
8	Settings	RTC time setting, restore default settings
9	Data export	USB/SD card data export
10	Log	View run log records

Note: A password is required when going from System Settings to the Calibration page.

6.2 Low-temperature Preservation

Figure 6-2 Low-temperature Preservation Settings Page

Low-temperature preservation means that after the instrument completes the main experimental program, it automatically lowers the module temperature to a specific temperature (usually 4°C) and maintains it, in order to prevent sample degradation.

- **Low-temp preservation switch:** ON/OFF toggle. When enabled, it automatically enters low-temperature preservation mode after the run ends.
- **Low-temp preservation temperature:** Setting range 4~20°C.
- **Low-temp preservation time:** Set the preservation duration (HH:MM:SS). It automatically exits low-temperature preservation mode when the time is reached.

Operating logic:

- **Auto handoff:** When the user-set main program (either "Manual Thermostat" or "Auto Program Control" mode) finishes running, if the low-temperature preservation function is enabled, the system seamlessly switches to the low-temperature preservation stage.
- **Timed mode:** If the user sets a "Low-temp preservation time" (e.g., 60 minutes), the instrument will maintain the low temperature for that duration, then automatically stop running and turn off temperature control (standby).
- **Always-on mode:** If no time is set (or time is set to 0/infinite), the instrument will maintain the low-temperature state until manually stopped by the user.

6.3 Power-on Preset

Figure 6-3 Power-on Preset Settings Page

Power-on preset means that after the instrument is powered on, it automatically executes a pre-set thermostatic program, allowing the instrument to quickly reach a commonly used standby temperature (such as 25°C or 37°C), reducing the user's waiting time before an experiment.

- **Power-on preset switch:** ON/OFF toggle.
- **Power-on preset temperature:** Setting range 4~100°C. When enabled, the device automatically starts controlling temperature to the preset temperature upon power-on.

Operating logic:

- **Auto start on power-up:** After the instrument is powered on and passes the self-test, if it detects that the "Power-on Preset" switch is enabled, it directly enters the preset thermostatic running state without user intervention.
- **Priority and interruption:** This mode has lower priority than the user's main program. During power-on preset operation, once the user starts a new main program (such as Program A), the power-on preset is immediately terminated, and the instrument gives priority to executing the user-specified main program.

6.4 Hot Lid Mode

Figure 6-4 Hot Lid Mode Settings Page

Hot lid mode is used to independently control the hot lid temperature, preventing condensation from forming on the inner wall of the sample lid during experiments such as PCR.

- **Hot lid mode switch:** ON/OFF toggle.
- **Hot lid temperature difference:** Set the difference between the hot lid temperature and the module temperature (5~20°C), not exceeding 110°C in total.
- **Note:** When the module temperature is below 20°C, the hot lid function is disabled.

Operating logic:

- **Independent control:** The hot lid's on/off is independent of the module temperature control. The user can separately set the hot lid's target temperature (typically set to 5~10°C above the reaction temperature, e.g., 105°C).
- **Anti-condensation mechanism:** When enabled, the hot lid continuously heats to the set value. By keeping the tube-top temperature higher than the reagent temperature inside the tube, it physically prevents water vapor from condensing on the tube cap, thereby ensuring constant reaction system concentration and the accuracy of optical detection (if any).
- **Associated behavior:** It is generally recommended to turn on the hot lid for preheating before the main program starts; in some modes, when the module temperature falls below a specific threshold (e.g., 15°C) entering low-temperature preservation, the hot lid may automatically turn off to assist cooling.

Note: When in hot lid mode, target temperature = current target temperature – hot lid temperature difference × 5%.

6.5 Log

Figure 6-5 Log Page

1. Every 5 lines display: serial number, start time (HH:MM:SS), end time (HH:MM:SS), run mode (Manual mode / Program control mode), and details.
2. Click the details icon to enter the log details page, which displays the detailed run information of that log entry.
3. Up to 100 log entries can be saved, paged through via "Previous Page" and "Next Page".
4. The log also has a power-loss save function.
5. Click the "Clear" button to clear all log entries. (Note: this process takes a certain amount of time.)

Note: The log details page is divided into Manual and Program Control pages.

6.6 Alarm Information

Figure 6-6 Alarm Information Page

1. Displays alarm information; up to 100 entries can be saved, paged through via "Previous Page" and "Next Page".
2. Alarm information also has a power-loss save function.
3. Click the "Clear" button to clear all alarm entries. (Note: this process takes a certain amount of time.)

Alarm types include:

Alarm type	Description
Temperature sensor fault	No valid temperature data detected during power-on self-test
Temperature abnormality	Temperature shows no change for an extended period during operation and has not reached the target temperature
Module not properly installed	Module ID not recognized
Communication fault	Serial screen communication abnormality
Memory fault	FRAM read/write abnormality
Hot lid not detected	Hot lid module not connected

6.7 Self-test

Figure 6-7 Self-test Page

Click the "Self-test" button, and the device will re-execute the power-on self-test process, checking whether the temperature sensor is working properly (whether valid temperature data is read within 10 seconds). After the self-test passes, it returns to the System Settings page. The fan speed is 50% during the self-test.

6.8 Calibration (Temperature Compensation)

The calibration function is used to compensate and calibrate the temperature measurement value, improving temperature control accuracy. The system supports 4 groups of temperature compensation values, corresponding to 4 temperature ranges:

Temperature range	Scope
Low-temperature zone	0~15°C (including 0°C)
Normal-temperature zone	15~55°C (including 15°C and 55°C)
Mid-temperature zone	55~85°C
High-temperature zone 1	85~100°C (including 85°C)
High-temperature zone 2	100~105°C (including 100°C)

Users can adjust the compensation value of the corresponding range according to the reading of a standard thermometer to make the displayed temperature more accurate.

- 5ml×8, 15ml×8, and 50ml×4 are large modules, using the large-module temperature compensation.
- 0.6ml×24, 1.5ml×24, and 2.0ml×24 well-plate modules are small modules, using the small-module temperature compensation.

6.9 Settings

The Settings page includes the following functions:

- **RTC time setting:** Set the device's real-time clock (year/month/day/hour/minute), providing accurate timestamps for log and alarm records.
- **Restore default settings:** Restore all parameters to factory default values.

6.10 Data Export

- **USB export:** Supports exporting run log data to a USB flash drive via the USB interface (EXCEL spreadsheet), facilitating data analysis on a computer.
- **SD card auto-save:** Supports periodic automatic saving of temperature data to an SD card, saving once every 5 minutes.

6.11 Product Information

Figure 6-8 Machine Information Page

Displays the basic product information:

Item	Content
Product name	B5 MAX Intelligent Metal Bath
Product model	B5 MAX
Software version	B5.MAX-V3.2.2
Screen version	V6.6.2
Time setting	Set the machine's year/month/day and hour/minute

7 Operation Instructions

7.1 Run Mode Introduction

The Metal Bath Heating & Cooling Instrument supports two main run modes:

1. **Manual mode:** Operation method - on the Manual Settings page, set the target temperature and hold time, then click the "Run" button to start. Stop method - click the "Stop" button, or it stops automatically when the hold time is reached.
2. **Program control mode:** Operation method - on the Program Edit page, select or edit a program, set the target temperature, hold time, and rate mode for each step, then return to the Program Control page and click "Run". Stop method - click the "Stop" button, or it stops automatically after all steps are completed.

7.2 Manual Mode

7.2.1 Preparation

- Confirm mode: Manual Settings page
- Set target temperature (4~100°C)
- Set hold time (optional; if set to 0, it runs continuously)
- Select rate mode (Basic / High-speed / Extreme)

7.2.2 Operation Method

Click "Run"; the device starts heating/cooling to the target temperature. The Run button lights up and the Stop button turns off.

After the temperature reaches the set value, the hold countdown starts automatically.

When the hold countdown ends, the buzzer beeps three times and the device stops automatically.

During operation, you can click the "Pause" button at any time to pause; click "Run" again to resume.

During operation, you can click the "Stop" button at any time to stop.

Notes:

1. If a temperature abnormality occurs after the device starts running, the device will stop automatically, display the alarm information on the alarm page, and record the alarm information.
2. Power-loss resume function: If an unexpected power outage occurs during operation, the device will automatically resume running from the breakpoint after power is restored.
3. If the module is not properly installed, the module model displays "Module not detected", and Run/Pause/Stop are all grayed out (including both manual and program control).
4. If the module is not properly installed, the module model displays "Module not detected", and Run/Pause/Stop are all grayed out (including both manual and program control).

5. During operation, the set temperature and set time input boxes cannot accept input, and the rate mode cannot be modified.

7.3 Program Control Mode

7.3.1 Preparation

- Confirm mode: Program Control page
- Select the program to run on the Program Edit page
- Confirm that the parameters of each step of the program (target temperature, hold time, rate mode) are correctly set

7.3.2 Operation Method

Click "Run"; the device begins executing the first step, heating/cooling to that step's target temperature.

After the current step temperature is reached, the hold countdown begins. When the countdown ends, it automatically switches to the next step, and the buzzer beeps once to indicate the step switch.

The current step number, target temperature, and hold countdown are updated in real time.

After all steps are completed, the buzzer beeps three times and the device stops automatically.

Notes:

1. During operation, click the "Stop" button to stop at any time.
2. When restarting after stopping, the current step, target temperature, and hold time will all be restored to the content corresponding to step 1 of the program.
3. If a temperature abnormality occurs after the device starts running, the device will stop automatically, display the alarm information on the alarm page, and record the alarm information.
4. Power-loss resume function: If an unexpected power outage occurs during operation, the device will automatically resume running from the breakpoint after power is restored (including the current step and remaining time).
5. The program name cannot be selected during program-controlled operation.
6. The lower heating plate is not allowed to exceed 108°C (physical limit; output may stop if exceeded).

8 Technical Specifications

Category	Parameter
Main control chip	STM32F407VETx (168MHz Cortex-M4F)
Operating system	FreeRTOS 10.x real-time operating system
Display screen	7-inch DGUS serial touch screen
Temperature sensor	PT1000 platinum resistor (4 channels)

ADC accuracy	AD7124 24-bit high-precision ADC
Memory	FRAM FM24V05 64KB (non-volatile on power loss)
Control algorithm	Dual closed-loop cascade PID (incremental + positional)
Communication protocol	Modbus, DGUS serial screen protocol
Temperature range	0°C ~ 105°C
Heating performance	Three heating modes: Full power: >5°C/min High: 5°C/min Medium: 2°C/min Heating time: ≤15min (room temp 25°C - 100°C)
Cooling performance	Three cooling modes: Full power: ≥2°C/min High: ≥1°C/min Medium: ≥0.5°C/min Cooling time: ≤15min (room temp 25°C - 4°C)
Adapted modules	Supports various 规格 test tubes (0.6, 1.5, 2, 5, 15, 50 mL) Multi-well plate modules (cell culture plates, ELISA plates, full-skirted microplates) Module replacement is simple, no tools required
Adapted lids	Standard regular module lid (precise temperature control) Optional hot lid (prevents low-temperature condensation, 1.5/2mL)
Program control programs	Up to 50 groups, each with up to 10 steps
Log capacity	100 run logs + 100 alarm records
Temperature curve	Up to 12,800 data points
Software scale	Approx. 21,000 lines of C code

Warm Tips

1. Before turning off the instrument, please stop the device first, wait for the instrument temperature to return to room temperature, and then turn off the power switch.
2. Do not restart the device immediately after turning off the power; improper operation will affect the service life of the instrument.

Scan to View the User Manual

(Use your mobile phone's WeChat or a browser to scan and view the instrument operation instructions)